INCH-POUND

MIL-T-53121
30 September 1991

MILITARY SPECIFICATION

TRUCK, LIFT, FORK, CLEAN BURNING DIESEL, FRONT/SIDE LOADING, 6000 POUND CAPACITY

This specification is approved for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for 6000 pound capacity, diesel engine driven, internal combustion engine, front/side loading, industrial depot type forklift trucks.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards for a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed inthe issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

GG-P-455	-	Plates and Foils, Photographic (Photo-Sensitive Anodized Aluminum).
VV-F-800	-	Fuel Oil, Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3930

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MILITARY

MIL-P-514	- Plate, Identification, Instruction, and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-1283	- Can, Gasoline, Military, 5-Gallon.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-E-10062	- Engine, Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive, and Artillery.
MIL-S-43926	- Suit, Chemical Protective.
MIL-G-43976	- Glove and Glove Set, Chemical Protective.
MIL-F-43987	- Footwear Cover, Chemical Protective (Overboots).
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative, Metric.
MIL-M-51282	- Mask, Chemical Biological Field, M17a.
MIL-H-51291	- Hood, Chemical Biological Mask M6a2.
MIL-V-81940	- Valve, Sampling and Bleed, Hydraulic, Type II Systems.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8, NATO-34 (JP-8) and NATO-35.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.
FED-STD-595	- Color (Requirements for Individual Color Chips) 3 x 5.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-162	- Materials Handling Equipment, Preparation For Shipment, Storage, Cyclic Maintenance, Routine Testing and Process.

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MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Materiel.
MIL-STD-1791	- Design for Internal Aerial Delivery in Fixed Wing Aircraft.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise, specified, the issues are those cited in the solicitation.

DRAWINGS

TACOM

12258941	- Connector, Receptacle, 54 Pin.
12258955	- Vehicle Test Card.

(Applications for copies should be addressed to Department of the Army, U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDCE, Warren, MI 48397-5000.)

REPORT

CR 82-588-003	- STE/ICE Design Guide for Vehicle Diagnostic Connector.

(Applications for copies should be addressed to Department of the Army, U.S. Army Tank-Automotive Command, ATTN: AMSTA-WST, Warren, MI 48397-5000.)

DEPARTMENT OF AGRICULTURE

Forest Service Standard 5100-1a

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 24401.)

DEPARTMENT OF LABOR (DOL)

29 CFR Title 29 Chapter XVII Part 1910 - Occupational Safety and Health Administration (OSHA).

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(Applications for copies should be addressed to a DOL Officer or the Superintendent of Documents, US Government Printing Office, Washington D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR, Part 86, Subpart B

Emission Regulation for 1977 and Later Model Year New Light Duty Vehicles and New Light Duty Trucks; Test Procedures (Particulate Tunnel and Sampling System) pp 402-465, 1 July 1983.

40 CFR, Parts 81 to 99, Subparts D and I

Protection of Environment.

(Applications for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B 93.28 - Method for Calibration of Liquid Automatic Particle Counters Using "AC" Fine Test Dust.
Z 26.1 - Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.

(Applications for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

B56.1 - Safety Standard for Powered Industrial Trucks.
B56.11.3 - Load Handling Symbols for Powered Industrial Trucks.
B56.11.4 - Forks and Fork Carriers for Powered Industrial Trucks.

(Applications for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook

J154 - Operator Enclosures Human Factor Design Considerations.

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J198 - Windshield Wiper Systems - Trucks, Buses, and Multipurpose Vehicles.
J209 - Instrument Face Design and Location for Construction and Industrial Equipment.
J297 - Operator Controls On Industrial Equipment.
J514 - Hydraulic Tube Fittings.
J516 - Hydraulic Hose Fittings.
J517 - Hydraulic Hose.
J518 - Hydraulic Flanged Tube, Pipe, and Hose Connections, 4-Bolt Split Flange Type.
J524 - Seamless Low Carbon Steel Tubing Annealed for Bending and Flaring.
J525 - Welded and Cold Drawn Low Carbon Steel Tubing Annealed for Bending and Flaring.
J534 - Lubrication Fittings.
J537 - Storage Batteries.
J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J598 - Sealed Lighting Units For Construction, Industrial and Forest Machinery.
J614 - Engine and Transmission Dipstick Marking.
J753 - Lubrication Chart - Construction and Industrial Machinery.
J759 - Lighting Identification Code.
J833 - Human Physical Dimensions.
J898 - Control Locations for Off-Road Work Machines.
J899 - Operator's Seat Dimensions for Off-Road Self-Propelled Work Machines.
J925 - Minimum Service Access Dimensions for Off-Road Machines.
J931 - Hydraulic Power Circuit Filtration.
J985 - Vision Factors Considerations in Rear View Mirror Design.
J1127 - Battery Cable.
J1165 - Reporting Cleanliness Levels of Hydraulic Fluids.
J1176 - External Leakage Classifications for Hydraulic Systems.
J1362 - Symbols for Controls, Indicators, and Tell-Tales for Off-Road, Self-Propelled Work Machines.

(Applications for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC. (NMFTA)

National Motor Freight Classification Rules

(Applications for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

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UNDERWRITERS LABORATORIES, INC. (UL)

UL 558 - Internal Combustion Engine - Powered Industrial Trucks.

(Applications for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules.

(Applications for copies should be addressed to Uniform Classification Committee, ATTN: Tariff Publishing Officer, RM 1106, 222 S. Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The truck shall be a clean burning diesel engine powered, front wheel drive, rear wheel steer, sit down rider, front/side loading, type DS truck, as described in UL 558. The truck shall have pneumatic or solid pneumatic (non-flat) tires with industrial tread. Truck capacity shall be 6000 pounds at 24 inch load center when operating as a counterbalanced truck in both front and side modes. A load carrying deck shall be provided for supporting the load when in the side carrying position.

3.1.1 Standard product. The truck shall be equipped with instruments, components, and accessories required for the specified operation of the truck. The truck shall be complete with all components that are standard with the manufacturer's products and in use by commercial users, whether stipulated herein or not, together with such accessories specified herein.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. The truck shall not contain any part that consists of any amount of asbestos material. No radioactive material shall be used in any part of the truck. Radioactive material is that material in which

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the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries.

3.3.1 Material deterioration prevention and control. The truck shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the truck may be exposed.

3.3.1.1 Dissimilar materials. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Rubber parts. All rubber parts shall be no more than 12 months old on date of acceptance by the Government.

3.3.2 Identification of materials and finishes. The contractor shall identify the specific material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the trucks may be newly fabricated from recovered materials to the maximum extent practicable, provided the trucks produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the trucks.

3.3.4 Screw threads. Screw threads shall be in accordance with FED-STD-H28.

3.4 Operating temperature. The engine shall start without external preheating within 5 minutes and the truck shall operate as specified herein within 15 minutes after the engine start in any ambient temperature required from -25 to +120 degrees Fahrenheit.

3.5 Performance. The truck shall meet the performance requirements as specified herein.

3.5.1 Lifting speed. The lifting speed of the forks with rated capacity load shall be not less than 60 feet per minute over the entire distance from ground level to maximum fork height.

3.5.2 Lowering speed. The lowering speed of the unloaded forks shall be not less than 40 feet per minute over the entire distance from maximum fork height to ground level. The lowering speed of the forks with rated capacity

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load shall be not more than 80 feet per minute over the entire distance from maximum fork height to ground level.

3.5.3 Free lift height. The free lift height, the height the forks will lift without increasing the specified collapsed mast height, shall be not less than the height of the load carry deck plus 5 inches.

3.5.4 Maximum fork height. The maximum fork height of the truck with rated capacity load, with the uprights fully extended, measured from the ground to the load carrying surface of the forks at the load center, shall be not less than 144 inches.

3.5.5 Drift. The lift assembly shall be capable of holding the rated load at maximum lift height for not less than 10 minutes with not more than 2 inches of vertical drift and not more than 1 degree of rotational drift.

3.5.6 Collapsed mast height. The collapsed mast height of a truck with rated capacity load measured from the ground to the top of the uprights in a vertical position shall not exceed 92 inches.

3.5.7 Upright tilt. When unloaded, the truck mast shall have the capability to forward tilt 3 degrees, plus or minus 1/2 degree, and a rearward of 6 degrees minimum.

3.5.8 Side carry position. When the rated capacity load is in the side carry position, the truck shall be capable of lifting and lowering the load to any position from ground level to maximum lift height. When a 43 inch wide load is supported in the side carry position on the load carrying deck, the maximum width of the truck and load shall not exceed 80 inches.

3.5.9 Pivot and traverse. Pivot and traverse functions shall be provided and shall be actuated by controls independent of each other. Pivot and traverse shall be accomplished by hydraulic means.

3.5.9.1 Pivot. With rated capacity load, the uprights shall pivot from the front carry position to the side carry position, 90 degrees to the right of the longitudinal centerline of the truck.

3.5.9.2 Traverse. With rated capacity load, the uprights shall traverse across a plane perpendicular to the longitudinal centerline of the truck.

3.5.9.3 Side clearance. The distance the vertical face of the fork extends beyond the right side of the truck when the uprights are pivoted 90 degrees and traversed to the extreme right shall be not less than 2 inches at any height from ground level to maximum lift height.

3.5.9.4 Load side-shift. With rated capacity load in the front carry position, the traverse function shall provide the capability to shift the load a minimum of 3 inches to either side of the truck's front carry position.

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3.5.9.5 Pick and carry. The truck shall be capable of picking and placing a rated capacity load (48 inches) out of a storage space not more than 60 inches wide, perpendicular to the longitudinal axis of the vehicle, when operating in a 60 inch aisle. The truck shall be able to enter the 60 inch aisle with the load by turning right or left from an intersecting aisle not wider than 190 inches.

3.5.10 Truck speed. While carrying the rated capacity load on a level surface, the truck shall be capable of attaining a speed of no less than 10 miles per hour both in the forward and reverse travel directions.

3.5.11 Truck acceleration. From a standing start, the truck carrying the rated capacity load on a level surface shall be capable of acceleration in the forward travel direction such as to cover a distance of 44 feet in not more than 8 seconds.

3.5.12 Slope ascension. The truck shall be capable of ascending a 20 percent grade on asphalt concrete or equivalent surface with and without rated capacity load. The truck shall be capable to accelerate from a dead stop on this slope when carrying the rated capacity load.

3.5.13 Truck underclearance. The truck with rated capacity load shall have sufficient underclearance to permit operation from one horizontal plane to another up or down a 20 percent slope. Minimum underclearance beneath the mast assembly, when in true vertical position, shall be not less than 8 inches.

3.5.14 Stability. The truck shall conform to "forward stacking", "forward travel", "lateral stacking", and "lateral travel" stability requirements of ASME B56.1.

3.5.15 Right angle turn. The truck, when carrying a rated capacity load, shall be capable of backing through a turn in either direction and returning to the same or a parallel position within dimension "A" of figure A-1 when demonstrated as specified herein. Dimension "A" shall not exceed 190 inches.

3.5.16 Curb climbing. The truck with and without rated capacity load shall be capable of ascending from a level ground over a 1-1/2 inch vertical curb onto another level surface. The truck shall climb over the curb from a standing start with the drive wheels against the curb.

3.5.17 Endurance/reliability. The truck shall have a mean time between failures (MTBF) of not less than 73 hours when tested in accordance with paragraph 4.6.2.

3.5.18 Maintenance ratio. The ratio of maintenance man hours to operating hours for all scheduled and unscheduled maintenance (exclusive of daily operator checks and services) during the first article test specified herein shall not exceed 0.19.

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3.6 Safety. The truck shall conform to the requirements of ASME B56.1, UL 558, CFR Title 29 Part 1910 in effect at the time of manufacture. The truck shall meet the requirements of UL 558 for DS type trucks. Individual trucks shall be labeled with safety rating. Safety rating marking shall be in accordance with UL 558.

3.6.1 Seatbelt. When specified (see 6.2), an adjustable, weather resistant seatbelt shall be provided with a corrosion resistant, metal-to-metal, quick release buckle. The seatbelt shall be furnished equipped with sleeves and retractors, or seat mounted.

3.7 Human factors engineering (HFE). The truck shall be operable and maintainable by a small woman through a large man as defined by SAE J833, who are wearing cold weather clothing, or when specified (see 6.2) wearing NBC protective gear (see 3.7.1). Controls and displays shall be labeled with universal symbols in accordance with SAE J1362. Controls shall be located in accordance with SAE J898 and SAE J297. Instrumentation shall be in accordance with SAE J209.

3.7.1 Nuclear-biological-chemical (NBC) protection. When specified (see 6.2), the truck shall be operable by personnel wearing the following complete protection ensemble.

a. Mask, Chemical-Biological, MIL-M-51282.
b. Hood, Chemical-Biological, MIL-H-51291.
c. Suit, Chemical Protective, MIL-S-43926.
d. Glove Set, Chemical Protective, MIL-G-43976.
e. Footwear Cover, Chemical Protective, MIL-F-43987.

The operator shall be capable of actuating all controls (switches, pedals, knobs, and handles) without interference caused by the ensemble that is detrimental to the operation of the truck. The floorboard in the operator's compartment shall not have exposed bolts or screws that can interfere with the operation of the truck while wearing protective footwear covers. The protective ensemble as listed above, will be loaned as Government-loaned equipment (GLE) to the contractor in order to permit performance of the test.

3.8 Maintainability. Provisions shall be made for adjustment, servicing, and replacement of all electrical assemblies and components, hydraulic system component, batters, belts, wear parts of lift and tilt mechanism brakes and components, wear parts of steering assembly, tires, wheels, lights, and horn. Test points identified for connection of the simplified test equipment for internal combustion engines to measure functions in the transducer kit (TK) mode (see 3.34.) shall be accessible without the removal of other major assemblies and installed attachments not normally removed. All fluid levels shall be accessible when the engine is at operating temperature without subjecting personnel to injuries or burns. If hand access openings are required to perform maintenance, the edges of each opening shall be smooth and the opening shall be provided with a removable hinge or cover. Dimensions of access opening shall be in accordance with SAE J925 for mittened hand. There

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shall not be interference by frame members or other obstructions when servicing or drawing oils and lubricants to or from any assembly or component. All reservoirs with removable drain plugs shall provide for complete drainage into a container. Engine, transmission, hydraulic system, steering system, and service brakes shall be furnished with dipsticks in accordance with SAE J614. In lieu of dipstick to check service brake and hydraulic system, visual inspection is acceptable.

3.8.1 Maintenance operations. Unless otherwise specified herein, each operation, in the following maintenance operations list, shall be accomplished by one person using common tools in not more than the time specified.

a. Remove, replace and adjust all engine driven belts - 1/2 hour.
b. Remove and replace alternator - 3/4 hour.
c. Remove and replace all filters, screens and strainers in hydraulic system - 1/2 hour.
d. Remove and replace fuel filter elements - 1/2 hour.
e. Drain engine lubricating oil, remove and replace oil filter elements and refill crankcase - 1/2 hour.
f. Remove and replace engine coolant system hoses - 1/2 hour.
g. Disconnect battery cables, remove and replace batteries and reconnect battery cables - 1 hour.
h. Drain torque converter oil if applicable and transmission oil, remove and replace all filter elements and strainers and refill converter and transmission - 1 hour.
i. Remove and replace starter - 1 hour.
j. Bleed and adjust brakes and refill master cylinder (two persons) - 3/4 hour.
k. Remove and replace headlight, floodlights and taillight bulbs - 1/4 hour each bulb.
l. Remove, replace and reinstall air filter - 1/2 hour.
m. Lubricate all lubrication fittings with two strokes per fitting using grease gun and extension if required - 1/2 hour.
n. Remove, read and replace each dipstick or conduct visual inspection - 2 minutes, each.
o. Drain water separator or replace filter element - 1/2 hour.

3.9 Lubrication. Lubricated means shall be provided for all moving parts that require lubrication, except where sealed permanent lubrication is provided. Each truck shall be serviced and run-in with military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-L-46167, MIL-G-10924, and MIL-B-46176. Initial servicing and run-in may be accomplished with lubricants standard to the contractor's commercial practice (except for braking system) if compatible with above military lubricants. (If commercial lubricants are utilized, the manufacturer will provide certification from the suppliers that commercial lubricants are compatible with military lubricants prior to the start of the first article test.) A lubrication data plate in accordance with SAE J753 shall be mounted on the truck and shall identify all military lubricants and appropriate temperature ranges to be utilized when servicing

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the truck. Grease lubrication fittings shall conform to SAE J534 and shall be in a protected but accessible position.

3.10 Transportability. The truck shall be capable of being transported by military or commercial trailers, trains, marine vessels, and aircraft. The truck shall be capable of withstanding the impact forces encountered in shipment, without damage or permanent deformation. The truck shall be equipped with tiedown or slinging provisions.

3.10.1 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3 for type II or III equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components.

3.10.2 Slinging provisions. The slinging provisions shall conform to MIL-STD-209, class 1 or 3, for type III equipment. The provisions shall enable the complete forklift to be lifted in the normal operating position. The provisions shall be located so that not less than one inch clearance is maintained between slings and all exterior parts and shall be fastened to members which shall withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provisions or its connecting structural components. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.10.1. All slinging/tiedown provisions shall be labeled "LIFT", "TIEDOWN", or "LIFT/TIEDOWN" as appropriate, in 1-inch high letters.

3.10.3 Air transportability. Unless otherwise specified (see 6.2), the truck shall meet the requirements of MIL-STD-1791 for air transport. The contractor shall have sufficient data available for the Government to confirm that the truck being supplied can be loaded and transported in C-130, C-141, and C5-A aircraft.

3.10.4 Rail transportability. Unless otherwise specified (see 6.2), the truck shall be rail transportable in CONUS and NATO countries without restrictions. The truck shall have a dimensional profile within the Gabarit International de Chargement (GIC) outline diagram when loaded on a 50-inch high rail car. The truck shall be capable of withstanding shock loads resulting from rail impact testing in accordance with MIL-STD-810, method 516.4, procedure VIII without failure, damage, or permanent deformation.

3.10.5 Highway transportability. Unless otherwise specified (see 6.2), the truck when loaded on a semitrailer/tractor, shall be within the highway legal limits of all States.

3.10.6 Marine transportability. Unless otherwise specified (see 6.2), the truck shall be transportable on container and breakbulk ships. The truck

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shall also be transportable on the LACX-30 and larger vessels of the Army tactical watercraft fleet.

3.11 Engine. The engine shall be a commercial type diesel engine. The engine furnished shall operate on fuel conforming to both VV-F-800, grade DF2 and MIL-T-83133. The engine shall be covered by an engine cowl or hood which can be easily opened or removed without tools. The engine shall be capable of operating with the applicable military lubricants specified in 3.9.

3.11.1 Diesel engine fuel system. The capacity of the fuel tank shall provide not less than 10 hours of continuous operation when tested in accordance with 4.6.2. The tank shall be located so that the tank can be filled from a 5-gallon can conforming to MIL-C-1283 by a person standing on the ground. The tank shall be equipped with a safety filler-cap assembly which shall incorporate a self-closing cap, screw-locking device for the cap, removable strainer, and shall provide for padlocking. The fuel tank shall be equipped with a drain plug at the lowest point of the tank so that the fuel can be completely drained. The drainplug shall be removable with hand tools without the removal of any other component. A fuel filter shall be provided and shall be accessible from above or from the side without removal of other components. If the fuel filter is located below the fuel tank, a fuel shut off valve shall be provided to assist in the maintenance or replacement of the fuel filter without draining the fuel from the tank. The fuel system shall include a fuel water separator located in the fuel line before the fuel filter, with provisions for draining sediment. A flame arrestor shall be installed in the fill pipe.

3.11.2 Cold weather starting system. Integral starting aids shall be provided to enable the engine to start under cold weather conditions down to -25 degrees Fahrenheit. If cold weather start system uses ether, then a measured shot system shall be employed that is not functional before cranking engine and not functional after the engine has started.

3.11.3 Crankcase ventilation system. A positive crankcase ventilation system shall be furnished to prevent crankcase vapor from entering the atmosphere.

3.11.4 Alternator. A 12 volt alternator of not less than 40-ampere capacity shall provide charge at engine low idle speed and maintain battery charge at full electric level. An alternator regulator shall be furnished. The regulator may be integral with the alternator.

3.11.5 Starting system. A device shall be furnished to prevent re-energizing the starter motor after the engine has been started. The truck shall be provided with an interlock in the starting system or other means to prevent energizing the starter motor, except when the transmission is in the neutral or park position. The starter shall be heavy duty and shall be in accordance with the engine manufacturer's recommendations. Starting system shall operate on 12 volts.

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3.11.6 Air cleaner system. A dry, replaceable cartridge type air cleaner shall be furnished. The cleaner shall be a multi-stage air cleaner and shall include a dust scavenger. The air cleaner shall be accessible without removal of other components when the engine cowl or hood is open. A restriction indicator shall be furnished to indicate when the replaceable filter element shall be changed, and shall be visible from the operator's seat.

3.11.7 Exhaust system. Muffler and exhaust piping shall conform to the requirements of UL 558 for DS type truck. A vertical exhaust shall be provided to a height near the top of the overhead guard and fitted with a rain cap to prevent entry of water. The open rain cap shall not exceed the overall height of the truck. Exhaust discharges shall be up and away from the operator's position.

3.11.8 Exhaust emission requirements for clean burn diesel engine. Exhaust emission constituents shall not exceed the limits set forth below when tested in accordance with applicable exhaust emission test paragraph of section 4. These emission tests shall be conducted with a fuel delivery system identical to and calibrated to give the same fuel delivery rate as the production engine used to demonstrate conformance to the endurance requirement (see 3.5.17) and to the cold start requirement (see 3.11.2). The diesel engine as provided by the manufacturer shall be capable of satisfying the exhaust emission requirements without the use of a catalytic convertor, trap oxidizer, water scrubber or other type of exhaust gas processor with the exception that a mechanical spark arrester which meets the requirements of the Forest Services Standard 5100-1a shall be provided. The spark arrestor shall be sized to meet the requirements for engine exhaust flow at rated engine speed in accordance with Forest Service Standard. However, the spark arrestor shall not be installed during emission testing.

3.11.8.1 Brake specific exhaust emission of hydrocarbons (HC), carbon monoxide(CO) and nitric oxides (NOx). When tested in accordance with 4.6.8.4.1, the brake specific exhaust emission of HC shall not exceed 0.5 g/bhp-hr; of CO shall not exceed 5.0 g/bhp-hr; and of NOx shall not exceed 6.0 g/bhp-hr, per Title 40 CFR.

3.11.8.2 Total suspended particulate (TSP). When tested in accordance with 4.6.8.4.2, the TSP shall not exceed 15 g/hr for 3, 4, 10 and 11 modes of the 13 mode federal test procedure.

3.11.8.3 Steady state smoke opacity (SSSO). When tested in accordance with 4.6.8.4.3, the SSSO shall not exceed 5 percent for any mode of the 13 mode federal test procedure excluding the high load modes of 5, 6, 8, and 9.

3.11.9 Oil filter. A replaceable full flow oil filter with emergency bypass shall be provided. Oil filter shall be located such that the filter element can be replaced without the removal of any other truck component.

3.11.10 Oil sampling valve. An oil sampling valve in accordance with MIL-V-81940 shall be provided on the engine. The valve shall be located in

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such a way as to insure that personnel shall not be exposed to danger when taking oil samples with the engine running. The location of the valve shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running. When specified (see 6.2), an oil sampling valve shall be provided for the transmission and hydraulic system. Sampling valve shall be labeled adjacent to the valve indicating the type of oil sampled (engine or transmission or hydraulic, whichever is appropriate).

3.11.11 Cooling system. When a liquid-cooled engine is furnished, the cooling system shall be of the closed pressure type, incorporating a radiator with a pressure cap, hoses, overflow container, thermostat, fan, and coolant circulating pump. The cooling system fluid shall be in a 50-50 mixture of antifreeze conforming to MIL-A-46153 and water. Accessible drain(s) shall be provided to completely drain the cooling system directly into containers.

3.12 Drive train components.

3.12.1 Transmission. The truck shall be equipped with a torque converter and either a power shifted (hydrodynamic) continuous drive type or a hydrostatic type transmission. Transmission shall provide for positive inching control of the truck. The transmission and torque convertor fluid shall be filtered by a full flow filter having replacement filters, without removal of other components. These filters shall conform to SAE J931.

3.12.2 Tires and tire loading. Tires shall be new and unused. Tires shall be pneumatic or solid pneumatic (non-flat) with industrial tread. Tire loadings shall not exceed the tire manufacturer's rating for this application. Tires shall be not more than 12 months old on date of acceptance by the Government.

3.13 Hydraulic system. The hydraulic system shall consist of all hydraulic components necessary for operation of the load handling means system and does not include brake, transmission, and steering system components. All hydraulic system component material shall be compatible with MIL-L-2104 and MIL-L-46167. The system shall provide for lowering of rated load at speed not to exceed 80 feet per minute in the event of failure or of damage to hydraulic hose supplying the lift cylinder. A straight-thread O-ring boss test point shall be located at the outlet of each pump or pressure inlet of each control valve for measuring pressure. Hydraulic hose(s) shall have a working pressure which is greater than the hydraulic system maximum relief valve setting. Hose reels shall be mounted such that they are within the plan outline of the truck. Positive means shall be furnished to prevent cavitation of the tilt cylinders. A means to determine fluid level in the reservoir (either dipstick or sight glass) shall be provided. A hydraulic system relief valve shall be provided to protect the hydraulic system from excessive pressure.

3.13.1 System filtration. System filter(s) shall be furnished in the reservoir, the suction line or the return line. The system filter shall meet the requirements of SAE J931. The particulate matter in the hydraulic fluid

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removed from the system shall not exceed the ISO code level of 19/16 in accordance with SAE J1165.

3.13.2 Hydraulic tubing, hoses and fittings. All hydraulic tubing, hoses and fittings shall meet the requirements of the applicable SAE standards J514, J516, J517, J518, J524, and J525 and shall be routed to preclude kinking, chafing, and twisting. All rubber hoses shall be no more than 12 months old on the date of acceptance by the Government.

3.13.3 Hydraulic pumps. The hydraulic pump(s) shall have a 1000 hour Omega rating which is less than the system filter Beta Ten rating.

3.13.4 Hydraulic control valves. Mounting of the valves shall be rigid and shall permit no vibration that may cause instability of the controls. All control valve spools shall be spring centered with sufficient force to return the levers to the center position and retain them there throughout all operations. Spools shall not bind under any operating conditions specified herein. All exposed control valve surfaces shall show no evidence of corrosion when operated as specified herein.

3.13.5 Hose reels. When hydraulic hose reels are used on the mast to maintain hose tension, sheave wheels, guide rollers, guide blocks, or other means shall be provided to prevent the hose from becoming entangled around and riding over the reels or reel flanges. Hose reels shall be mounted such that they are within the plan outline of the truck for all positions of the forks.

3.13.6 External leakage. After completing tests specified in 4.6 the degree of external hydraulic system leakage shall not exceed a class 4 for dust free conditions and a class 4 D for dusty conditions of SAE J1176. External hydraulic system leakage shall not exceed a class 3 for production trucks under normal operation.

3.14 Load backrest. A metal load backrest, removable without disturbing any other component and using only common tools, shall be provided. There shall be no protruding bolts or appendages beyond the side plane of the load backrest. The load backrest in conjunction with fork and carriage design shall provide a vertical rear guard at least 48 inches high measured from load carrying surfaces of the forks and equal to the width of the carriage. Spacing between vertical members shall not be more than 6 inches. The load backrest shall have width and capacity and strength sufficient to prevent a rated load 48 inches high or less from forcing the backrest against the upright when the mast is at maximum rearward tilt. The backrest shall be capable of withstanding without permanent deformation a horizontal static load equal to the rated load uniformly distributed over the area, from the top carrying surface of the forks to the top of the backrest for not less than 3 minutes.

3.15 Uprights and carriage assembly. Uprights shall be high visibility telescopic roller type. Rollers shall be the permanently lubricated-for-life type. The fork carrier shall conform to ASME B56.11.4. The fork carrier

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shall be free of obstruction to allow the forks to slide across the entire width of the carrier. A cut out shall be provided in the fork carrier so that the lower hook of the fork will fit through the slot and each fork shall be removable without removing the load backrest.

3.16 Forks. Forks shall conform to ASME B56.11.4. Forks shall be replaceable without removing the load backrest. Fork spacing shall provide for adjustment of the fork spread from a minimum of 12 inches to a maximum of not less than 45 inches. The fork spacing shall be measured from the outside to outside edge of the forks. Fork thickness shall not exceed 1/2 inch at the tips and the end taper shall be not less than 14 inches long. Taper shall be on the underside of the forks. Fork size shall be 2-3/8 inches maximum thickness, 6 inches maximum width, and 42 inches (plus 0 inch, minus 1/2 inch) long. The top of the fork tips shall be chamfered or rounded to break corners and provide smooth edges.

3.17 Load carrying deck. A load carrying deck shall be provided for supporting long loads when in the side carry position. Long loads are missile containers that measure from 116 inches up to 240 inches in length.

3.18 Steering. Power steering shall be furnished. In the event of power failure, steering shall require less than 50 pounds of pull at the rim to turn the steering wheel.

3.19 Brakes.

3.19.1 Service brakes. Hydraulically actuated service brakes shall be provided. Brakes shall be self adjusting. The brakes shall be of non-asbestos materials that do not degrade performance or increase maintenance. The brake system performance shall be in accordance with ASME B56.1.

3.19.2 Parking brake. A parking brake shall be provided to engage a minimum of two wheels when the parking brake is applied. The braking capability shall be sufficient to hold the truck with and without rated capacity load while ascending slope specified in 3.5.12. No more than 35 pounds of force shall be required to apply the parking brake.

3.19.3 Brake fluid. Brake system shall be compatible and filled with MIL-B-46176 brake fluid. All brake system components utilized within the truck brake system shall be compatible with silicone brake fluid. A metallic tag shall be attached to the brake master cylinder stating "Use Silicone Brake Fluid Only."

3.20 Towing device. A pin type towing device shall be provided in the center rear of the truck, recessed into the counterweight.

3.21 Controls and instrumentation. In addition to the requirements of 3.7, all load handling control symbolic markings shall be in accordance with ASME B56.11.3.

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3.21.1 Instruments. An ammeter or indicator light, engine oil pressure gauge or indicator light, transmission temperature gauge, and fuel gauge shall be provided. All gauges and indicator lights shall be identified, designed and located in accordance with SAE J209. Trucks equipped with a liquid cooled engine shall be equipped with an engine temperature gauge or indicator light. All gauges shall be weather resistant and internally lighted.

3.21.2 Hour meter. An electrically operated hour meter which registers the number of engine operating hours up to a minimum of 9999 hours shall be provided. The hour meter shall be accessible for reading without removing any component of the truck.

3.22 Electrical system. Electrical system voltage shall be 12 volts DC, negative ground.

3.22.1 Battery. Trucks shall be furnished with a minimum of one 12 volt battery conforming to SAE J537. Battery cables conforming to SAE J1127 shall be furnished with insulated covers. The positive terminal shall be colored red and labeled "+". The negative terminal shall be colored black and labeled "-". A pictorial battery connection diagram shall be provided. The diagram shall be located such that it is viewable by the maintainer during battery maintenance.

3.22.2 Ignition switch. A key operated ignition switch shall be furnished. It shall have not less than two positions. Two ignition keys shall be furnished with each truck. All keys shall be identical for every truck. When specified (see 6.2), a keyless ignition switch shall be furnished.

3.22.3 Floodlamps. Each truck shall be equipped with 4 sealed beam floodlamps which shall conform to SAE J598 Par 46, Trade No. 4478. The floodlamps shall be shock mounted in elastomer ring housings with 1 floodlamp mounted on the left front of the overhead guard, 1 floodlamp mounted on the right of the mast, and 2 floodlamps mounted on the rear of the overhead guard. Directional adjustment range of each floodlamp shall be a minimum of 45 degrees above and below the horizontal plane. Horizontal adjustment range of each floodlamp shall be a minimum of 90 degrees. One on-off switch for the 2 front mounted floodlamps, and one on-off switch for the 2 rear mounted floodlamps shall be provided within access of the seated operator.

3.22.4 Taillight. A minimum of one combination stoplight and tail-lamp shall be provided and shall conform to SAE J759. The tail-lamp shall be mounted on the rear within the plan outline of the truck, and shall be protected against damage by a guard. The tail-lamp shall come on when the switch for the front mounted floodlamps is turned on.

3.23 Operator's overhead guard. A removable overhead guard shall be furnished which meets applicable requirements of ASME B56.1. The overhead guard shall be removable with hand tools. Guard height, measured from ground level to the top side of the guard, shall not exceed 92 inches. Guard design shall not interfere with the operation of the truck, nor with normal movements

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of the operator when entering, leaving or operating the truck. The overhead guard shall remain firmly secured to its support brackets under all operating conditions specified herein.

3.24 Seat. The seat shall conform to the requirements of SAE J899 except that armrests are not required.

3.25 Personnel cab. When specified (see 6.2), the truck shall be equipped with an insulated, weathertight personnel cab. The interior cab dimensions shall be in accordance with SAE J154. The cab shall inherently resist torsional strain and withstand all operational shock and vibration under the operating conditions specified herein. The cab shall be constructed of metal. The cab shall be equipped with one fully opening or removable door. The door shall have a safety latch, opening stop and hold-open catch. All hardware shall be of the heavy-duty or industrial type, resistant to corrosion. The cab shall be equipped with not less than two ventilation fans not less than 6 inches in diameter. Fans shall be safely guarded and directionally adjustable to direct airflow towards all glazing. Fan(s) shall have "on-off" switch(es) which are labeled "ON" and "OFF" and are within the seated functional reach of a small woman operator. The cab design shall not interfere with the vertical exhaust pipe (see 3.11.7).

3.25.1 Glazing. The cab shall provide the operator with all around visibility. The total glazed area shall be not less than 4000 square inches of tinted windows consisting of windshield, sliding or roll-down door windows, rear window, roof window, and other windows as determined by cab design. All glazing shall conform to ANSI Z 26.1, type AS1 in the windshield and roof window and type AS1 or AS2 in other windows. Fixed glazing shall be sealed in waterproof stress-absorbent molding, and movable windows shall be sealed on extra heavy type, pile lined channels.

3.25.2 Interior. The interior of the cab shall be fitted with insulation, retarding both heat loss and noise intrusion. The insulation and framing shall be covered with an interior skin and headliner, providing a smooth, cleanable protective cover. Interior dome lights shall be provided to enable the operator to read all control labels under conditions of darkness.

3.25.3 Windshield and window wipers. The cab shall be equipped with at least three electrically powered wiper assemblies to clear the windshield, the rear window and the roof window in accordance with the performance requirements of SAE J198. Self-parking features shall be provided. Wiper arms and blades shall be of the commercial heavy-duty type. The wiper arms shall provide tension on the glazed surface of at least 1 ounce for each inch of wiper arm length. The operator controls shall be installed on or adjacent to the instrument panel.

3.25.4 Heater. A recirculating air type heater shall be furnished. The heater shall have a rating of at least 18,000 Btu/hour. The heater controls shall be near the operator and shall provide for variable selection of blower speeds.

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3.26 Rearview mirror. When specified (see 6.2), the truck shall be equipped with a rearview mirror (with mounting arm) mounted on the right hand side within the operator's field of view as described in SAE J985.

3.27 Fire extinguisher. When specified (see 6.2), a 5 pound carbon dioxide (CO_2) fire extinguisher shall be mounted on each truck in an easily accessible location.

3.28 Horn. The truck shall be equipped with an electric horn with a pushbutton mounted in the center of the steering wheel. The horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant to prevent entry of moisture when operated or stored outdoors under all weather conditions.

3.29 Noise limits. The noise produced by the truck shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474 5.2, 5.3, and 5.4, when tested in accordance with 4.6.14. The provisions of MIL-STD-1474, 4.2 and 4.3, shall be provided if and only if MIL-STD-1474, 5.1.1.3 requirements have been met and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity.

3.30. Treatment and painting. The parts of the truck normally painted shall be cleaned, treated and painted in accordance with the contractor's commercial practice. When specified (see 6.2), the truck shall be treated and painted in accordance with MIL-T-704, type A, and the final color shall be in accordance with color 13538 of FED-STD-595. When specified (see 6.2), the truck shall be treated and painted in accordance with MIL-T-704, type F for ferrous metals and G for nonferrous metals, and the final color shall be 33538 aircraft yellow or 34094 green 383.

3.31 Identification marking. Unless otherwise specified (see 6.2), each truck shall be identified in accordance with MIL-STD-130 with an identification plate conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material which shall be located in a visible and permanent location on the truck. Unless otherwise specified (see 6.2), a shipping data plate and instruction plates/labels shall conform to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1). The shipping data plate shall indicate the silhouette of the truck in the transport position indicating the center of gravity and the location and capacity of the slinging and tiedown provisions. The instructions plates/labels shall include warnings, cautions and diagrams describing any special or important procedures to be followed in operating, towing or servicing the truck. All plates shall be securely attached to the truck with bolts, rivets or screws and shall be furnished and mounted by the contractor. An instruction plate indicating "12-volt system" shall be provided and shall be located in a position visible to the operator and service personnel.

3.31.1 Truck marking. When the truck is provided with contractor commercial color, contractor commercial markings shall be provided. When yellow or CARC paint is provided (see 3.30) markings shall be black.

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Characters shall be block-type capital letters and Arabic numerals. Location and character size shall be as shown in table I.

TABLE I. Marking location and size.

Information	Example	Location	Minimum Character Size (Ht.)
Agency	US	On each side and rear	1-1/2 inches
Registration No.	10D315	On each side and rear	1-1/2 inches
Capacity	60001b. cap.	On each side	2 inches
Safety	No Smoking	On Rear of mast	1 inch
	No Riders	On rear of mast	1 inch
Tire Pressure (Pneumatic only)	TP100 psi	On frame above each tire	1 inch
Type Fuel	Diesel	Near fuel tank filler	1 inch
Lifting/Tiedown Points	Lift/Tie-down	Near lift/tiedown points	1 inch

3.32 Electromagnetic radiation. The truck shall not exceed the limits of electromagnetic radiation specified in SAE J551.

3.33 Nuclear survivability. When specified (see 6.2), schematic, electrical and mechanical diagrams of the truck detailed to the component level (i.e., resistors, capacitors, filters, transistors, materials), shall be available from the contractor to assist the Government in performing the nuclear survivability evaluation. When specified (see 6.2), a truck shall be furnished to the Government and shall be evaluated by the Government to determine necessary protection devices/measures for nuclear weapons effects.

3.34 Diagnostic connector assembly (DCA) measurement capabilities. Unless otherwise specified (see 6.2), the truck shall incorporate an easily accessible DCA located in or near the operator's compartment for interface with the Simplified Test Equipment for Internal Combustion Engines Reprogrammable (STE/ICE-R) test equipment. The DCA shall be in accordance with TACOM drawing no. 12258941 and shall be labeled "Diagnostic Connector Assembly". All requirements for DCA shall be in accordance with the STE/ICE-R design guide for vehicle Diagnostic Connector Assemblies, report no. CR-82-588-003, revision 1. The DCA shall have the capabilities for measuring the minimum functions defined in table 2-2 of report no. CR-82-588-003, revision 1. Additional test parameters to be monitored in either the DCA or transducer kit (TK) mode shall be as follows: fuel return pressure, engine oil pressure, engine oil filter pressure drop, starter solenoid voltage, and transmission oil pressure. Test points identified to monitor functions in the TK mode shall be accessible without the removal of major assemblies and installed attachments. Those additional test parameters which are inaccessible for monitoring without the removal of other components in the TK

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mode shall be monitored in the DCA mode. Fittings/adapters required for interface with the STE/ICE-R test set when monitoring in the TK mode, shall be permanently installed on the truck to insure that adapters in the STE/ICE-R test set readily mate with the TK mode test point locations. A separate wiring harness shall be provided for the DCA and shall include all wiring and necessary hardware to perform required capabilities. A fuel shut-off method shall be provided for conducting compression unbalance checks. The contractor shall also provide STE/ICE-R vehicle test cards in the format specified in TACOM drawing 12258955 addressing both DCA and TK measurements, and shall incorporate the test cards into the truck technical manuals.

3.35 Government-loaned equipment. Unless otherwise specified (see 6.2), the following items shall be loaned by the Government (see 6.5):

a. NBC protective ensemble:

1. Chemical-biological mask, NSN 4240-00-926-4200.
2. Chemical-biological hood, NSN 4240-00-999-0420.
3. Chemical protective suit, NSN 8415-00-407-1062.
4. Chemical protective glove set, NSN 8415-01-033-3519.
5. Chemical protective footwear, NSN 8430-01-021-5978.

b. STE/ICE-R

3.36 Workmanship. Each truck shall have no evidence of cracks, dents, scratches, burrs, sharp edges, loose parts, missing fasteners, distortions, spatter, or foreign material. Workmanship of the truck should, in no case, be less than that of comparable products furnished to the commercial market.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to requirements prescribed.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known

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defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections, evaluations, and demonstrations. The inspection requirements specified herein are classified as follows:

a. First article truck inspection (see 4.4).
b. Quality conformance inspection (see 4.5).
c. Inspection comparison (see 4.8)
d. Inspection of packaging (see 4.9).

4.3 Inspection Conditions. Unless otherwise specified herein, all examinations and tests shall be performed at the ambient temperature and climatic conditions existing at the place of examination and test. Only that maintenance established by the contractor and submitted as the maintenance schedule (see 4.4) prior to commencement of the test shall be performed during the test.

4.4 First article truck inspection. Prior to production the manufacturer shall furnish one or more trucks (see 6.2) produced from production tooling for first article evaluation and test (see 6.3). The truck and its components shall be compared with this document, published literature, and calculations and test data to verify compliance with this specification. The truck shall then be subjected to tests and examination as marked in column 1, table II and column 1, table III. When the contractor can provide verifiable data demonstrating compliance to any or all parts of the specified inspection and testing, these inspections and tests may be waived by the Government. Prior to examination and tests, the following shall be performed.

a. Service the truck with the oils, greases, and fuel specified herein and designated for use in the ambient temperature at which the test will be conducted.
b. Conduct production run-in as specified in 4.7.
c. Set the hydraulic system relief pressure as specified by manufacturer and record. Any change made to this shall constitute failure of all tests.
d. Furnish a schedule of maintenance to be followed during all testing of the first-produced trucks. A list of special tools required to perform this maintenance shall also be provided.

4.4.1 First article examination. Prior to testing, the first-produced truck(s) shall be examined for the defects marked "X" in column 1 of table II. Presence of one or more defects shall constitute failure of this test.

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TABLE II. Examination schedule.

First Article	Quality Conformance	Defect	Requirement Paragraph
1	2	3	4
X	-	101. Material not as specified.	3.3
X	-	102. Material not resistant to corrosion or treated to be resistant to corrosion and deterioration for the applicable storage and operating environment.	3.3.1
X	-	103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1.1
X	X	104. Rubber parts not as specified.	3.3.1.2
X	X	105. Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.2
X	X	106. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the trucks.	3.3.3
X	-	107. Screw threads not as specified.	3.3.4
X	X	108. Safety not as specified.	3.6
X	X	109. Seatbelt not as specified.	3.6.1
X	X	110. Human factors engineering not as specified.	3.7
X	-	111. Maintainability not as specified.	3.8
X	X	112. Lubrication provisions, lubricants not as specified.	3.9
X	X	113. Engine not as specified.	3.11
X	X	114. Fuel system not as specified.	3.11.1
X	X	115. Crankcase ventilation not as specified.	3.11.3
X	X	116. Alternator not as specified.	3.11.4
X	X	117. Starting system not as specified.	3.11.5
X	X	118. Air cleaner system not as specified.	3.11.6
X	X	119. Exhaust system not as specified.	3.11.7
X	X	120. Oil filter not as specified.	3.11.9
X	X	121. Oil sampling valve not as specified.	3.11.10
X	X	122. Cooling system not as specified.	3.11.11
X	X	123. Transmission not as specified.	3.12.1
X	X	124. Tires not as specified.	3.12.2
X	X	125. Hydraulic system not as specified.	3.13
X	X	126. System filtration not as specified.	3.13.1

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TABLE II. Examination schedule - continued.

First Article	Quality Conformance	Defect	Requirement Paragraph
1	2	3	4
X	X	127. Hydraulic tubing, hoses, and fittings not as specified.	3.13.2
X	X	128. Hydraulic pump not as specified.	3.13.3
X	X	129. Hydraulic control valves not as specified.	3.13.4
X	X	130. Hose reels not as specified.	3.13.5
X	X	131. External leakage not as specified.	3.13.6
X	X	132. Load backrest not as specified.	3.14
X	X	133. Uprights and carriage assembly not as specified.	3.15
X	X	134. Forks not as specified.	3.16
X	X	135. Load carrying deck not as specified.	3.17
X	-	136. Steering system not as specified.	3.18
X	-	137. Service brakes not as specified.	3.19.1
X	-	138. Parking brake not as specified.	3.19.2
X	X	139. Brake fluid not as specified.	3.19.3
X	X	140. Towing device not as specified.	3.20
		141. Controls and instrumentation not as specified.	3.21
X	X	142. Instruments not as specified.	3.21.1
X	X	143. Hour meter not as specified.	3.21.2
X	X	144. Electrical system not as specified.	3.22
X	X	145. Battery not as specified.	3.22.1
X	X	146. Ignition switch not as specified.	3.22.2
X	X	147. Floodlamps not as specified.	3.22.3
X	X	148. Taillight(s) not as specified.	3.22.4
X	X	149. Operators overhead guard not as specified.	3.23
X	X	150. Seat not as specified.	3.24
X	X	151. Personnel cab not as specified.	3.25
X	-	152. Glazing not as specified.	3.25.1
X	X	153. Cab interior not as specified.	3.25.2
X	X	154. Windshield and window wipers not as specified.	3.25.3
X	X	155. Heater not as specified.	3.25.4
X	X	156. Rearview mirrors not as specified.	3.26
X	X	157. Fire extinguisher not as specified.	3.27
X	X	158. Horn not as specified.	3.28
X	X	159. Treatment and painting not as specified.	3.30
X	X	160. Identification marking not as specified.	3.31

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TABLE II. Examination schedule - continued.

First Article	Quality Conformance	Defect	Requirement Paragraph
1	2	3	4
X	X	161. Truck marking not as specified.	3.31.1
X	X	162. Workmanship not as specified.	3.36

4.4.2 First article tests. Upon successful completion of examinations specified in 4.4.1, the truck(s) and components shall be subjected to the test marked "X" in column 1 of table III. The DCA interface test (see 4.6.17.1) shall be conducted as the first test for the first article truck(s). Following completion of all previously specified tests for one of the first article trucks, drain and flush the entire fuel system and refill the fuel system with fuel conforming to MIL-T-83133 and subject the truck to the travel speed test (see 4.6.1.9), the slope ascension test (see 4.6.1.11), and the endurance/reliability test (see 4.6.2). Failure of any of these tests or nonconformance to 3.11 shall constitute failure of this test. Acceptance of a first-produced truck shall not exclude the remaining trucks from the quality conformance inspections and acceptance provisions specified in section 4.

TABLE III. Test schedules.

First Article 1	Quality Conformance 2	Test 3	Test Paragraph 4	Requirement paragraph 5
X	-	Lifting and lowering speed.	4.6.1.1	3.5.1 and 3.5.2
X	-	Free lift height.	4.6.1.2	3.5.3
X	-	Maximum fork height.	4.6.1.3	3.5.4
X	-	Drift.	4.6.1.4	3.5.5
X	-	Collapsed mast height.	4.6.1.5	3.5.6
X	-	Upright tilt.	4.6.1.6	3.5.7
X	-	Side carry position.	4.6.1.7	3.5.8
X	-	Pivot.	4.6.1.8.1	3.5.9.1
X	-	Traverse.	4.6.1.8.2	3.5.9.2
X	-	Side clearance.	4.6.1.8.3	3.5.9.3
X	-	Load side-shift.	4.6.1.8.4	3.5.9.4
X	-	Pick and carry.	4.6.1.8.5	3.5.9.5
X	-	Truck speed.	4.6.1.9	3.5.10
X	-	Acceleration.	4.6.1.10	3.5.11
X	-	Slope ascension.	4.6.1.11	3.5.12
X	-	Underclearance.	4.6.1.12	3.5.13
X	-	Stability.	4.6.1.13	3.5.14

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TABLE III. Test schedules - Continued,

First Article 1	Quality Conformance 2	Test 3	Test Paragraph 4	Requirement paragraph 5
X	-	Right angle turn.	4.6.1.14	3.5.15
X	-	Curb climbing.	4.6.1.15	3.5.16
X	-	Endurance.	4.6.2	3.5.17
X	-	Maintenance ratio.	4.6.2.3	3.5.18
X	-	Nuclear-biological-chemical.	4.6.3	3.7.1
X	-	Maintenance operations.	4.6.4	3.8.1
X	-	Slinging and tiedown.	4.6.5.1	3.10.1 and 3.10.2
X	-	Air transportability.	4.6.5.2	3.10.3
X	-	Rail transportability.	4.6.5.3	3.10.4
X	-	Highway transportability.	4.6.5.4	3.10.5
X	-	Marine transportability.	4.6.5.5	3.10.6
X	-	Cold start test.	4.6.6	3.11.2
X	-	Starting system test.	4.6.7	3.11.5
X	-	General emission test. HC, CO and NOx.	4.6.8.4.1	3.11.8.1
X	-	Total suspended particulate.	4.6.8.4.2	3.11.8.2
X	-	Steady state smoke opacity.	4.6.8.4.3	3.11.8.3
X	-	Tire loading.	4.6.9	3.12.2
X	X	Hydraulic system cleanliness.	4.6.10	3.13.1
X	-	Fork spacing.	4.6.11	3.16
X	-	Steering test.	4.6.12	3.18
X	-	Braking test.	4.6.13	3.19.1 and 3.19.2
X	-	Noise level measurement.	4.6.14	3.29
X	-	Electromagnetic radiation.	4.6.15	3.32
X	-	Nuclear survivability.	4.6.16	3.33
X	X	Diagnostic connector assembly capability.	4.6.17	3.34
X	X	DCA-STE/ICE-R interface.	4.6.17.1	3.34

4.4.3 First article examination and test failure. Failure of a first article truck to meet any requirement specified herein during and as a result of the examination and test specified in 4.4.1 through 4.4.2 shall be cause for rejection of the first-article truck(s) and for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the manufacturer at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the first article inspection will be considered prima facie evidence that all trucks

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accepted prior to the completion of the first article inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Quality conformance examination. Each truck shall be examined for the defects marked "x" in column 2 of table II. Presence of one or more defects shall be cause for rejection.

4.5.2 Quality conformance tests. After successful completion of all examinations specified in 4.5.1, each truck shall be subjected to production run-in as specified in 4.7, and tested as specified in column 2 of table III. Failure of any test shall be cause for rejection.

4.5.3 Sampling.

4.5.3.1 Unit of product. A unit of product is defined as a fully-serviced truck.

4.5.3.2 Lot. A lot shall consist of not more than 25 units of product. Lots shall be formulated from trucks which have been successfully completed the individual examination and tests as specified in 4.5.1 and 4.5.2.

4.5.3.3 Sample tests. One sample shall be selected from each lot and shall be subjected to the tests specified in 4.6.1.1, 4.6.1.2, 4.6.1.3, 4.6.1.4, 4.6.1.6, 4.6.1.8.1, 4.6.1.8.2, 4.6.1.8.4, 4.6.1.9, 4.6.1.15, and 4.6.14. Any sample failing any test shall be cause for rejection of the lot.

4.6 Demonstrations and tests.

4.6.1 Performance.

4.6.1.1 Lifting speed and lowering speed. Measure and record distance in inches from floor to top surface of forks in fully lowered position. Record time in seconds required to raise rated capacity load to maximum fork height using a timing device. Record the difference between maximum fork height and the height of the forks in the lowered position. Record time in seconds to lower rated capacity load at maximum speed (full open valve) to 3 or 4 foot height, load to be abruptly stopped at the 3 to 4 foot height. Repeat the raising and lowering three times. Raise rated capacity load to maximum height and simulate hose failure to demonstrate conformance to 3.13. Remove load and raise empty forks to maximum lift height. Record time in seconds required to lower forks to lowered position. Nonconformance to 3.5.1 or 3.5.2 shall constitute failure of this test.

4.6.1.2 Free lift height. With no load and mast in the true vertical position, measure and record the free lift height. Nonconformance to 3.5.3 shall constitute failure of this test.

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4.6.1.3 Maximum fork height. With rated capacity load on the forks and mast in the true vertical position, measure and record maximum fork height. Nonconformance to 3.5.4 shall constitute failure of this test.

4.6.1.4 Drift. Start the truck and allow the hydraulic system oil to reach normal operating temperature. Position the truck on a level surface and raise the rated load to the maximum lift height with uprights in a true vertical position. Immediately mark a reference point on upright. After 10 minutes record vertical and rotational drift. Nonconformance to 3.5.5 shall constitute failure of this test.

4.6.1.5 Collapsed mast height. With no load and mast in the true vertical position, measure and record the collapsed mast height. Nonconformance to 3.5.6 shall constitute failure of this test.

4.6.1.6 Upright tilt. Place truck without load on flat level surface. Raise forks to an elevation of 2 feet. Tilt mast to full forward position and record angle of tilt, measured from the front surface of the outer mast channel. Tilt mast to full rearward position and record angle of tilt measured from the same surface. Nonconformance to 3.5.7 shall constitute failure of this test.

4.6.1.7 Side carry position. With rated capacity load on the forks and in the side carry position, adjust the uprights for true vertical position. Measure and record the maximum lift height. Place a 43 inch wide load on the forks and support on the load carry deck. Measure the overall width of truck and load in this position. Nonconformance with 3.5.8 shall constitute failure of this test.

4.6.1.8 Pivot and traverse assembly test.

4.6.1.8.1 Pivot. With no load on the forks and the mast in the front carry position, lift the forks to a height of no less than 50 inches. Adjust the upright tilt to achieve true vertical position. This shall be the starting position for this test. Engage the pivot control in order to motion the forks from front carry to a position 90 degrees to the longitudinal centerline of the truck. The pivot control shall be disengaged momentarily in order to stop the motion at the halfway point, and engaged again for completing the motion. After the pivot motion is completed, measure and record the angle of the fork location with respect to the centerline of the truck. Return the forks to the starting position. Perform the procedure 10 times with no load on the forks and 10 times with rated capacity load. Nonconformance to 3.5.9.1 shall constitute failure of this test.

4.6.1.8.2 Traverse assembly. With rated capacity load on the forks and the mast in the front carry position, lift the forks to a height of no less than 50 inches. Adjust the upright tilt to achieve true vertical position. This shall be the starting position for this test. Engage the traverse control in order to motion the load along a plane perpendicular to the truck centerline, from the extreme left to the extreme right of the travel distance. Perform

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this procedure 10 times, alternating between front carry fork position, side carry fork position and an intermediate position between front carry and side carry positions. Nonconformance to 3.5.9.2 shall constitute failure of this test.

4.6.1.8.3 Side clearance. With no load on the forks and the mast in the front carry position, lift the forks to a height of no less than 50 inches. Adjust the upright tilt to achieve true vertical position. This shall be the starting position for this test. Engage the pivot control in order to motion the forks from front carry to a position 90 degrees to the longitudinal centerline of the truck. Engage the traverse control in order to motion the forks to the extreme right position. Measure and record the clearance between the vertical face of the forks and the right side of the truck. Return the forks to the starting position. Perform the procedure 5 times with no load on the forks and 5 times with rated capacity load. Nonconformance to 3.5.9.3 shall constitute failure of this test.

4.6.1.8.4 Load side-shift. With rated capacity load on the forks and the mast in the front carry position, lift the forks to a height of no less than 50 inches. Adjust the upright tilt to achieve true vertical position. This shall be the starting position for this test. Engage the traverse control in order to move the load to the left and right of the truck front carry position. Measure and record the distance the load shifts each side of the front carry position. Perform this procedure 5 times. Nonconformance to 3.5.9.4 shall constitute failure of this test.

4.6.1.8.5 Pick and carry. Simulate a pallet storage rack with a 60 inch wide operating aisle perpendicular to the rack. Place a rated capacity load cube in a storage space not more than 60 inches wide. The intersecting aisle to the operating aisle shall not be more than 190 inches wide. Drive the truck from the intersecting aisle into the operating aisle and, using the pivot and traverse functions, retrieve the load from the storage space. With the load in the front carry position, exit the operating aisle in the reverse direction. With the load in the front carry position, drive the truck from the intersecting aisle into the operating aisle and, using the pivot and traverse functions, place the load in the storage space. Nonconformance with 3.5.9.5 shall constitute failure of this test.

4.6.1.9 Truck speed. Operate the truck at governed speed on a level surface with rated capacity load in front carry position. Truck shall attain speed specified in both directions. The speed shall be the average of three runs in each direction. Inability of the truck to attain the minimum speed as specified in 3.5.10, failure of any component, over-heating, malfunction, or leakage of fluid shall constitute failure of this test.

4.6.1.10 Acceleration test. Place the truck with rated capacity load in the front carry position on a level surface. From a standing start, accelerate the truck in the forward direction and measure the time required to cover a distance of 44 feet. Nonconformance to the requirements of 3.5.11 shall constitute failure of this test.

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4.6.1.11 Slope ascension. Drive truck forward with rated capacity load in front carry position along a horizontal approach and onto a 20 percent ramp or slope and bring to a complete stop. Engage parking brake to determine if brake will hold the truck on slope. Release parking brake. Start from a dead stop and proceed up the ramp to a horizontal surface. Remove rated capacity load and repeat above test. Nonconformance to 3.5.12 shall constitute failure of this test.

4.6.1.12 Truck underclearance. Drive the truck forward with rated capacity load in the front carry position along a horizontal approach and onto a 20 percent ramp or slope and proceed up the ramp to a horizontal surface. Proceed travelling down the slope in reverse direction back to the starting position. Repeat the above test without load. With no load on the truck, and the mast in true vertical position, measure and record truck underclearance under the mast assembly. Nonconformance to the requirements of 3.5.13 shall constitute failure of this test.

4.6.1.13 Stability. Perform applicable stability test specified in ASME B56.1. Nonconformance to 3.5.14 shall constitute failure of this test.

4.6.1.14 Right angle turn. The truck shall be tested in accordance with test method no. 1. Nonconformance to 3.5.15 shall be cause for rejection.

4.6.1.15 Curb climbing. Prepare a level surface which is 1-1/2 inches above level ground. The edge of the elevated surface shall form a 1-1/2 inch high curb. The truck shall be placed with the drive wheels against the curb. From a standing start accelerate the truck in order to climb over the curb and onto the elevated surface. This test shall be performed five times each with the rated capacity load in the front carry position, with the rated capacity load in the side carry position traversed to the extreme right, and without a rated capacity load. Nonconformance to 3.5.16 shall be cause for rejection.

4.6.2 Endurance/reliability testing. The truck shall be operated for a minimum of 490 hours over a course in accordance with test method no. 2. Hours specified here shall be productive clock hours excluding fueling, servicing and downtime. Only routine scheduled maintenance as prescribed by the technical manual is allowed. Record the reason and amount of time involved for each truck stoppage or malfunction, including refueling, during this test. During this test, the truck shall operate a minimum of 10 hours without refueling as specified in 3.11.1.

Accept/Reject Criteria.

- Accept with zero (0) endurance failures, reject otherwise.
- Accept with four (4) or less reliability failures. Reject with five (5) or more reliability failures.
- Relevant failures shall be determined by authorized Government representative. Nonconformance shall constitute failure of this test.

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4.6.2.1 Endurance failure. Any incident resulting in component failure of the engine, transmission, transfer case, differential, hydraulic pump drive axle, steer axle or permanent deformation of the truck chassis or upright assembly shall constitute an endurance failure.

4.6.2.2 Reliability failure. A reliability failure is defined as any malfunction which cannot be repaired in 10 minutes by the operator using on board tools and equipment which causes cessation, degradation, or inability to commence specified operations; or any other unscheduled maintenance action which requires more than one hour to complete. All such malfunctions shall be considered failures except for:

1. Installation damage.
2. Accident or mishandling.
3. Failure of the test facility or test peculiar instrumentation.
4. Equipment failures caused by an externally applied overstress condition, in excess of approved test requirements.
5. Normal operating adjustments (nonfailures) specified in the approved equipment instructions.
6. Secondary failures within the equipment which are directly caused by nonrelevant or relevant primary failures. The secondary failures must be proven to be dependent on the primary failure.
7. Failures caused by human errors.
8. Non-repetitive replacement of light bulbs, fuses, tires and tightening of fasteners.

Any impending or active malfunction detected during test which constitutes a safety hazard to personnel shall be considered a reliability failure.

4.6.2.3 Maintenance ratio. The maintenance ratio shall be calculated during the endurance/reliability test. Nonconformance to 3.5.18 shall constitute failure of this test.

4.6.3 Nuclear-biological-chemical (NBC) protection. The forklift shall be tested to determine conformance with 3.7.1. Nonconformance to 3.7.1 shall constitute failure of this test.

4.6.4 Maintenance operations. Perform the maintenance operations as listed in 3.8.1. Nonconformance to 3.8.1 shall constitute failure of this test.

4.6.5 Transportability test.

4.6.5.1 Slinging and tiedown provisions test. The slinging and tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.10.1 and 3.10.2. Inability to meet the requirements of 3.10.1 and 3.10.2 shall constitute failure of this test.

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4.6.5.2 Air transportability test. The truck shall be evaluated to determine conformance to 3.10.3. Inability to meet the requirements of 3.10.3 shall constitute failure of this test.

4.6.5.3 Rail transportability test. The truck shall be evaluated to determine conformance to the rail transport dimensional requirements of 3.10.4. The first article truck shall be tested in accordance with MIL-STD-810, method 516.4, procedure VIII to prove conformance with 3.10.4. Inability to meet the requirements of 3.10.4 shall constitute failure of this test.

4.6.5.4 Highway transportability test. The truck shall be evaluated to determine conformance to 3.10.5. Inability to meet the requirements of 3.10.5 shall constitute failure of this test.

4.6.5.5 Marine transportability test. The truck shall be evaluated to determine conformance to 3.10.6. Inability to meet the requirements of 3.10.6 shall constitute failure of this test.

4.6.6 Cold start test. Test the forklift in accordance with test method no. 3. One or more of the following shall constitute failure of this test:

a. Damage to any part of the truck due to storage at -25 °F (see 3.4).
b. Engine does not start within 5 minutes after beginning the first attempt (see 3.4).
c. Engine does not run smoothly within 15 minutes after starting without unnatural or continued control manipulation (see 3.4).
d. Instruments and gauges do not operate.
e. Battery charging system does not show positive charge.
f. Fluid leakage.

4.6.7 Starting system. Start the truck with the transmission in park or neutral, energizing the starter switch. Allow the engine to run for a minimum of 15 seconds. Place the transmission in gear while the engine is running and attempt to energize the starter motor by operating the starter switch. Nonconformance to 3.11.5 shall constitute failure of this test.

4.6.8 Diesel exhaust emissions.

4.6.8.1 Diesel exhaust emission tests. Diesel exhaust emission testing shall not commence until the contractor has demonstrated conformance to the endurance and cold start requirements.

4.6.8.2 Selection/verification preparation of diesel engine for exhaust emission tests. Diesel exhaust emissions test shall be conducted on two engines selected randomly from the production stock of engines. Each engine shall be equipped for dynamometer testing with production accessories (air cleaner, alternator, fan, etc.) as tested in the endurance and cold start tests. The fuel delivery system of the two test engines shall be identical and their adjustment and fuel delivery rates shall be determined and verified

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to be within 1 percent of those of the engine used in the endurance and cold start tests. If the fuel delivery systems are not identical and adjusted and calibrated to be within 1 percent of each other, this shall be cause to prohibit commencement of further emission testing until corrected. Each of the test engines shall then be run-in for at least 80 hours. During the run-in the engines shall be subjected to an even distribution of the loads and speeds from the 13 modes of the federal test procedure. After the run-in is complete, the horsepower of each of the emission test engines shall be determined. Each engine shall be capable of developing horsepower within 5 percent of the rated horsepower at rated speed as published by the engine manufacturer for this application. Failure of the test engines to develop sufficient horsepower until corrected shall be cause to prohibit further testing.

4.6.8.3 General emission test conditions. Each emission test specified herein shall be performed two times with each emission test engine and the results averaged to obtain the value which is compared to appropriate specification requirements. All emission testing shall be conducted using a D2 diesel control fuel as specified by 40 CFR Parts 81-99. The 80 hour run-in may be conducted with the contractor's choice of diesel fuel.

4.6.8.4 Emission test procedures.

4.6.8.4.1 Emission test procedures for HC, CO and NOx. Determine the emissions rate of HC, CO and NOx in accordance with the steady-state 13 mode federal test procedure (FTP), for new diesel heavy duty engines as specified by Title 40 Code of Federal Regulations Part 86, Subpart D. Brake specific values after being averaged in accordance with 4.6.8.3, of more than 0.5 g/bhp-hr HC, of 5.0 g/bhp-hr CO, and 6.0 g/bhp-hr NOx as specified in 3.11.8.1 shall constitute failure of this test.

4.6.8.4.2 Total suspended particulate (TSP) emission test procedure. The TSP emission measurement system shall be in accordance with procedures outlined by the Federal Register Volume 40, Part 86. Measurement shall be made, however, with the engine at steady state modes taken from the 13 mode federal test procedure. Testing for TSP 3, 4, 10, and 11 modes of the 13 mode test shall be conducted. Average TSP emission rates exceeding 15g/hr at any mode as specified in 3.11.8.2 shall constitute failure of this test.

4.6.8.4.3 Steady state smoke opacity test and procedures. Steady state smoke opacity levels, shall be determined using the smoke measurement system specified by Title 40 Code of Federal Regulations Subpart I for new diesel heavy duty engines. Instead of the 3 modes specified in subpart I, the engine modes to be tested shall be taken from the steady state 13 mode federal test procedures (FTP) as specified by Title 40, Code of Federal Regulations Subpart D for new gasoline fueled and diesel heavy duty engines. The high load modes 5, 6, 8, and 9 shall be excluded from the smoke test. Measurement of smoke opacity at all of the remaining of the 13 mode FTP test shall be conducted. Average smoke opacity values exceeding 5 percent at any mode as specified in 3.11.8.3 shall constitute failure of this test.

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4.6.9 Tire loading. Determine the weight supported by the steer (trail) wheels with and without a rated load on the forks by driving the wheels onto a platform scale or utilizing two axle scales. Determine the weight supported by driving (load) wheels in the same manner as the steer wheels. Nonconformance to 3.12.2 shall constitute failure of this test.

4.6.10 Hydraulic system cleanliness. Test each truck in accordance with test method No. 4. Continue cleanup procedures until the average of not less than three consecutive counts conforms to 3.13.1.

4.6.11 Fork spacing. Determine the minimum and maximum fork spacing adjustment. Nonconformance to 3.16 shall constitute failure of this test.

4.6.12 Steering test. With the engine stopped and the truck travelling in the forward direction at a speed not greater than 2 miles per hour, measure the force tangential to the steering wheel required to turn the rim of the steering wheel. Nonconformance to 3.18 shall constitute failure of this test.

4.6.13 Braking test. Perform applicable braking test specified in ASME B56.1. Nonconformance to 3.19.1 or 3.19.2 shall constitute failure of this test.

4.6.14 Noise level measurement. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated in MIL-STD-1474, figure 11. Noise levels shall be measured at the operator's position while the truck is performing the endurance/reliability test specified in 4.6.2. The peak dB(A) noise level experienced for five different one hour periods shall be reported. The level shall be reported in dB(A) and octave bands. A description of the operations being performed when the peak occurred shall also be reported (i.e. Test Method 2, procedure 2.g).

4.6.15 Electromagnetic radiation. Test the truck in accordance with SAE J551. As a minimum, the electromagnetic radiation report shall include results of all testing and a list of components used to reduce the electromagnetic radiation. Nonconformance to 3.32 shall constitute failure of this test.

4.6.16 Nuclear survivability. The Government will evaluate the schematic, electrical and mechanical diagrams to determine possible problem areas that may be encountered on the truck. A production truck shall be subjected to nuclear survivability testing by the Government at a Government facility. (See 3.33).

4.6.17 DCA capability. The contractor will perform a 100 percent visual inspection of all DCA wiring harnesses prior to installation on the truck to insure wires and connections are free of error. After installation on the truck, test the DCA harness (reference CR-82-588-003 revision 1) on each vehicle with a DCA tester prior to interface with STE/ICE-R. Nonconformance to 3.34 shall constitute failure of the DCA.

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4.6.17.1 DCA-STE/ICE-R interfaces. Upon completion of the tests specified in 4.6.17, test the DCA for all measurements specified in 3.34 using STE/ICE-R. After completion of all first article testing, this test shall be repeated and the results compared to the first test (4.4.2). Nonconformance to 3.34 or failure of any componentry or wiring in the DCA harness which results in an inability to obtain the measurements previously recorded with the STE/ICE-R, shall constitute failure of this test.

4.7 Production run-in. Each truck carrying a rated capacity load shall be subjected to a production run-in and servicing as specified herein. Inability to complete the run-in, overheating, failure of any components, malfunction of any control, or evidence of leakage of fluids in excess of class 3 in accordance with SAE J1176 shall be cause for rejection of the truck by the Government until such deficiencies are corrected.

4.7.1 Travel. Demonstrate starting, ascending and descending a 20 percent ramp, steering, inching and high speed travel. Operate all lights, horn, and brake system not less than two times during this demonstration.

4.7.2 Mast assembly. Demonstrate ability to lift rated capacity load to full lift height at full lift speed and lower load at full opened valve. Demonstrate ability to side shift rated capacity load throughout full range of side shift. Demonstrate ability to feather controls of all hydraulic functions. Each operation shall be performed at least three times.

4.7.3 Servicing and adjustment. Unless otherwise specified (see 6.2), prior to delivery, the manufacturer shall service and adjust each truck for customer operational use, including at least the following:

a. Adjustment of mast assembly.
b. Adjustment of engine, transmission, and power train.
c. Adjustment of brake system.
d. Adjustment of steering.
e. Continuity check of the electrical system, including filling and charging of battery unless otherwise specified in the preservation and packing requirement.
f. Check torque on all wheel lug nuts and proper inflation of tires, if required.
g. Service the cooling system with an antifreeze water solution, 50-50 percent by volume. Antifreeze shall conform to MIL-A-46153.
h. Check fluid levels and fill to required level if necessary.

4.8 Inspection comparison. The Government may select trucks at any time during the contract production period and subject the trucks to the examination specified in column 1 of table II and to the test specified in column 1 of table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article trucks. In addition to any test specified as part of the

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inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.8.1 Inspection failure. Failure of an inspection comparison truck to meet any requirement specified herein during and as a result of the examination as specified in 4.8 shall be cause for rejection of the inspection comparison truck(s) and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all trucks accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.9 Inspection of packaging. The preservation, packing, and marking of each truck shall be inspected as specified in appendix B of MIL-STD-162.

5. PACKAGING

5.1 Preservation. Each complete truck shall be preserved in accordance with the level A, B, or C requirements for type I (mobile) of MIL-STD-162 as specified (see 6.2). For all levels, the engine cooling system shall be preserved using the water/antifreeze procedure of MIL-E-10062.

5.2 Packing. Packing shall be A, B, or C as specified (see 6.2).

5.2.1 Level A or B. Each complete truck shall be packed in accordance with the level A or B requirements, type I (mobile) of MIL-STD-162.

5.2.2 Level C. The complete truck shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking. Marking shall be in accordance with MIL-STD-162.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The trucks describe herein are intended for stacking, unstacking, and moving palletized ammunition and missile containers in and around ammunition storage buildings. The truck is intended for operation over paved, semi-prepared, and other hard surfaces for short distances.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1. and 2.2).
c. When first article test is not required (see 3.2).
d. When a seatbelt is required (3.6.1).
e. When NBC operability is required (see 3.7).
f. Air transportability requirements (see 3.10.3).
g. Rail transportability requirements (see 3.10.4).
h. Highway transportability requirements (see 3.10.5).
i. Marine transportability requirements (see 3.10.6).
j. When an oil sampling valve shall be provided for the transmission and the hydraulic system (see 3.11.10).
k. When a keyless ignition switch shall be provided (see 3.22.2).
l. When a weather-tight personnel cab is required (see 3.25).
m. When the truck shall be equipped with a rearview mirror (see 3.26).
n. When the truck shall be equipped with a fire extinguisher (see 3.27).
o. When the truck shall be treated and painted in accordance with MIL-T-704, type A with final color 13538 or FED-STD-595 (see 3.30).
p. When the truck shall be treated and painted in accordance with MIL-T-704, type F and G with final color 33538 or 34094 of FED-STD-595 (see 3.30).
q. When each truck shall be identified with an identification plate other than that specified (see 3.31).
r. When each truck shall be identified with a shipping data plate and instruction plate labels other than that specified (see 3.31).
s. When schematics, electrical and mechanical diagrams of the truck for a nuclear survivability evaluation are required (see 3.33).
t. When a truck shall be furnished for a nuclear survivability evaluation (see 3.33).
u. When DCA is not required (see 3.34).
v. When Government loaned equipment is not required (see 3.35).
w. Number of trucks required for the first article (see 4.4).
x. When servicing and adjustment of the truck before delivery is not required (see 4.7.3).
y. Level of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the items should be a first article truck. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the first article. When the contractor can provide verifiable data demonstrating compliance to any or all parts of the specified inspection and testing, these inspections and test may be waived by the government.

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6.3.1 Incident report. When the contractor conducts the demonstration and test specified herein, a written report within 24 hours of any malfunction or failure during the conduct of the demonstration and test is required. As a minimum, the report shall describe components and parts affected, test and operating conditions, date of incident, hour meter reading, how detected, and description of incident and corrective action taken. A minimum of one original will be furnished the pertinent contracting officer specified in the contract. Distribution of copies will be identified in the contract.

6.4 Disposability. One or more of the following methods shall be used to accomplish disposal of lubricating oils, antifreeze, batteries, and paint: reuse, recycling, baling, sanitary landfill, composting, incineration, pyrolysis, or sea disposal.

6.5 Government-loaned property. The contracting officer should arrange to furnish the property listed in 3.7.1.

6.6 Noise levels. When the required maximum noise limit(s) are established to be technically infeasible and beyond the state-of-the-art for the item/system being procured, the contracting officer may request additional information on noise sources. As such, documentation shall be furnished to the procuring activity for consideration of whether or not increase of the required limit(s) is justified and how noise levels may best be minimized. Documentation shall contain technically defensible data including technically supported design considerations, technically supported design recommendations for noise reduction, and technically supported predictions of the resultant noise levels. Clearly stated, convincing, and technically/fiscally supported trade-off analysis of noise control benefit against other design requirements such as weight, access, etc., shall be provided.

6.7 Transportability report. The contracting officer should arrange for the contractor to prepare a Transportability Report in accordance with Data Item Description (DID) DI-PACK-80880 or AR 70-47 to be furnished to the Military Traffic Management Command, Transportation Engineering Agency (MTMC-TEA) prior to the delivery of the first article truck(s).

6.8 Definitions.

6.8.1 Load center. The horizontal longitudinal distance from the intersection of the horizontal load carrying surfaces and vertical load engaging faces of the forks to the center of gravity of the load.

6.8.2 Rated capacity load. Rated capacity load shall be established with a load equivalent to an unrestricted cube with overall dimensions twice the load center dimensions, and whose center of gravity is located at the geometrical center of the cube.

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6.9 Subject term (key word) listing.

Diesel engine, clean burning
Front/side loading
Truck, forklift

Custodians:
Army - ME
Navy - SA

Preparing activity:
Army - ME

Review activity:
Army - SM

Project 3930-0661

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TEST METHOD NO. 1

RIGHT ANGLE TURN

1. TEST COURSE:

   a. Level paved surface.
   b. Test course in accordance with "course layout" (see figure A-1).

2. TEST APPARATUS:

   a. Tape measure.
   b. Movable vertical barrier.
   c. Inclinometer.

3. TEST PROCEDURE:

   a. Truck is placed with the most forward edge of centered rated load coincident to line "B". (See figure A-1.) When forks are longer than rated load, the fork tips shall touch line "B" and the fork length is included in the "A" dimension. Mast in vertical position.
   b. Pallet is raised 6 inches above floor.
   c. Position movable vertical barrier within 4 inches of counterweight of truck.
   d. Turn steer wheels to the extreme left position and back truck through 90-degree turn.
   e. Measure the perpendicular distance from line "B" to the position assumed by the movable barrier. This is dimension "A".
   f. Return truck to initial position.
   g. Turn steer wheels to extreme right and repeat a through e.

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Test Method No. 1

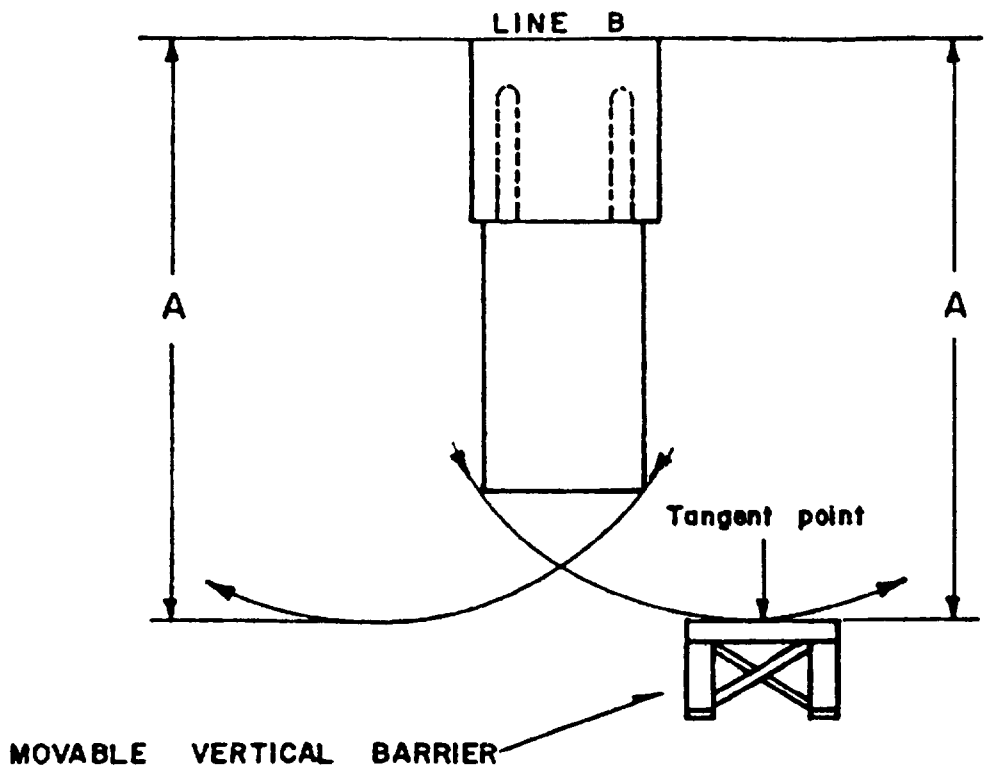


Figure A-1. COURSE LAYOUT

X-2308

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TEST METHOD NO. 2

ENDURANCE/RELIABILITY

1. TEST COURSE:

a. The test course shall be a level, improved surface laid out in accordance with figure A-2 or equivalent length course if a 200 foot diameter pad is not available. Station 1 and station 3 shall be platforms that are not less than 8 feet in height and capable of supporting the rated load. Station 2 shall be at ground level.
b. Obstacles shall be in accordance with figure A-3 and placed so that the nearest obstacle is not less than 10 feet from station 1.

2. TEST PROCEDURE:

a. Initially, place rated loads on platforms at stations 1 and 3.
b. Pick up load from station 1; lower the rated load to 12 inches above ground level; position the mast to maximum rearward tilt; back up truck, being sure all wheels travel over obstacles; travel forward and deposit load at station 2. Before reaching station 2, stop the truck and pivot the mast with load 90°, then pivot mast back to front carry position.
c. Disengage load at station 2; back out; then travel forward to station 3. Prior to engaging load at station 3, raise mast to maximum lift height to activate relief valve. Lower the forks and engage load.
d. Pick up load at station 3; lower the rated load to 12 inches above ground level; position the mast to maximum rearward tilt; back out; travel forward and deposit load at station 1, being sure all wheels travel over the obstacles.
e. Disengage load at station 1; back out, being sure all wheels travel over the obstacles; then travel forward to station 2.
f. Pickup load at station 2; raise the load to 12 inches above ground level; position the mast to maximum rearward tilt; back out; then travel forward with the load; before reaching station 3, stop the truck and pivot the mast 90°, then pivot the mast to front carry position; deposit load at station 3.
g. Steps 2(b) through 2(f) shall constitute one cycle. For 50 percent of the cycles, station 2 and station 3 shall be interchanged so that they are in clockwise position rather than in their initial counterclockwise position. Stations shall be changed at the end of each 8-hour operating day.

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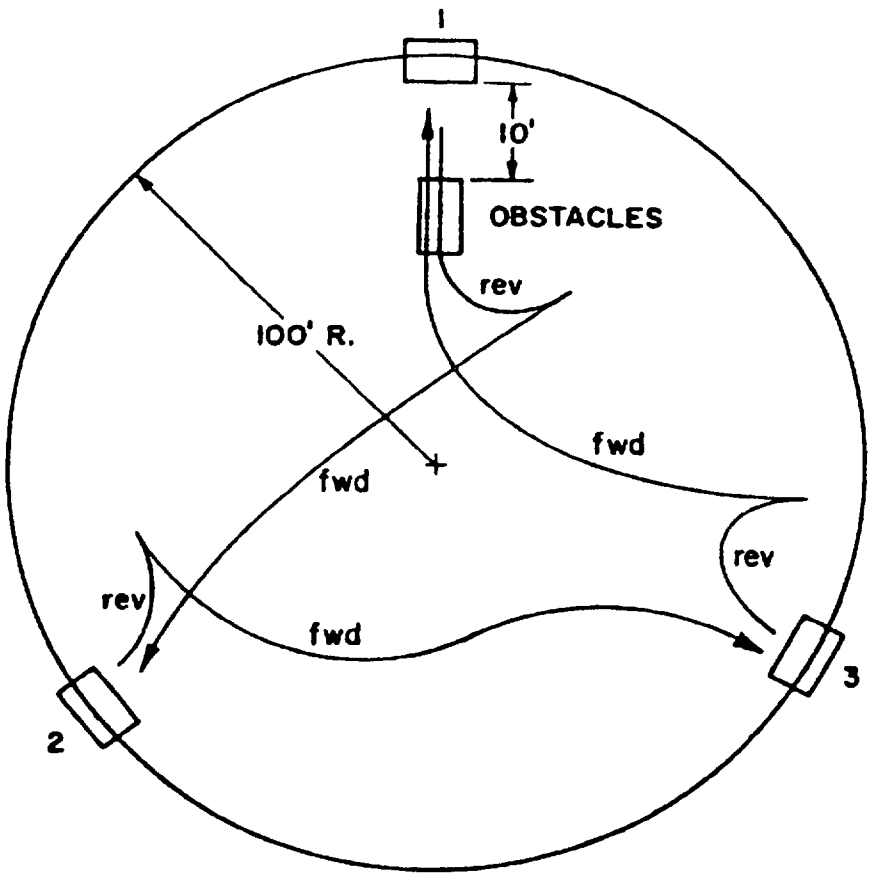
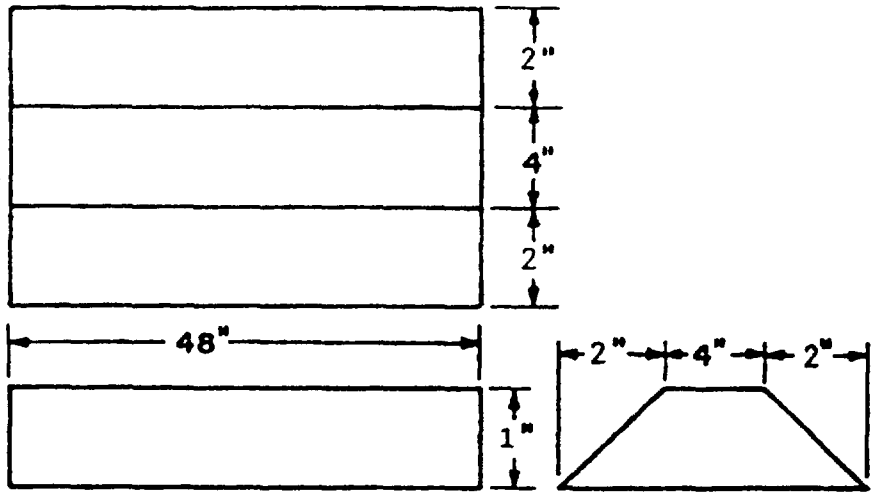


Figure A-2 Endurance/Reliability Course

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LAYOUT OF OBSTACLE COURSE

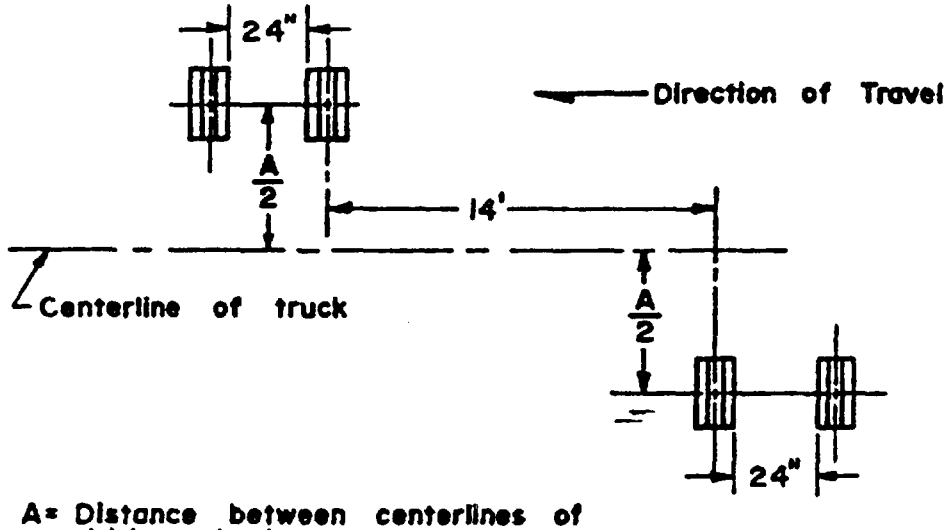


Figure A-3 DETAILS OF OBSTACLE CONSTRUCTION

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TEST METHOD NO. 3

LOW-TEMPERATURE TEST (MINUS 25°F)

1. TEST APPARATUS:

   a. Cold chamber.
   b. Arctic clothing.
   c. Thermocouples with potentiometer(s) or other temperature measuring device.
   d. Battery hydrometer.
   e. Wattmeter or ammeter.
   f. Voltmeter.

2. TEST PROCEDURE:

   a. Service the forklift with applicable lubricants for -25 °F, operation including engine oil, transmission oil, drivetrain lubricants, hydraulic oil and antifreeze.
   b. Check the specific gravity of battery electrolyte to insure full charge.
   c. Cold soak the forklift to attain a stabilized temperature no warmer than -25 °F. During and after this period, examine the forklift for damage. Stabilization is attained when the engine coolant, fuel, lubricating oil, and battery temperatures reach -25 °F, as indicated by thermocouples located as follows:

      (1) In the two center cells of each battery between the plates and at mid-depth of the electrolyte (unless sealed maintenance free batteries are furnished.)
      (2) In the volumetric center of the hydraulic system oil reservoir.
      (3) In the engine lubricating oil gallery and in the oil sump.
      (4) In the engine coolant in the engine-cooling-jacket discharge pipe or air cooling exhaust chamber of air cooled engines.
      (5) In the chamber at a horizontal angle of 45 degrees from the sides and ends of the truck, not less than 3 feet from each corner of the truck at engine-crankshaft level.

   d. Maintain the -25 °F, ±5 °F, temperature throughout the entire test.
   e. One operator wearing arctic clothing, including mittens, shall perform all priming and starting operations.
   f. Substitute or supplementary batteries or other electrical cranking energy shall not be used.
   g. Glow plugs working off the cranking batteries and auxiliary fluid priming systems which are permanently mounted on the forklift and normally used in starting the engine may be used.
   h. Start the forklift. Starting should be accomplished within 5 minutes of the initial attempt. Smooth engine running without unnatural or continued control manipulation, and without use of primers and priming fluids, should be obtained within 15 minutes after starting.

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i. When started, run engine 15 minutes. After completion of the 15 minute operation, operate the engine as follows:

(1) 10 minutes idling.
(2) 10 minutes half throttle.
(3) 5 minutes full throttle.

Perform the 25-minute cycle three times.

j. Maintain and record temperatures prior to engine starting, immediately after engine starting, and at 10 minute intervals during warmup and cycling, at the following (but not limited to) locations on the forklift:

(1) Battery (unless sealed maintenance free batteries are furnished).
(2) Hydraulic tank.
(3) Engine oil.
(4) Engine coolant.
(5) Chamber air.

k. During engine cycling, and while wearing arctic mittens and boots, the operator shall perform each of the following:

(1) Engage and disengage power train to the wheels in all gears.
(2) Operate through two complete movement cycles all brakes, shafts and cylinders (as permitted by space limitations in chamber).
(3) Mount to and dismount from the operator's position on the forklift.

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TEST METHOD NO. 4

HYDRAULIC SYSTEM CLEANLINESS

1. TEST APPARATUS:

   a. Liquid automatic particle counter.

2. TEST PROCEDURE:

   a. Connect the hydraulic lines at the cylinders to bypass those cylinders whose capacity is less than 1/2 of the connecting line volume.
   b. Run the engine at no load governed speed pumping the oil through each circuit for not less than 5 minutes. Perform the particle count as described in "d" below.
   c. When applicable, connect each line to its related component. Operate each function of the system for 5 minutes continuously through full stroke and at maximum rates. Perform the particle count as described in "d" below.
   d. The procedure for determining contamination shall be as follows:

      (1) Contamination levels shall be determined by particle counts. The particle counter used for evaluation of the samples shall be calibrated in accordance with ANSI Recommended Standard Method for Calibration of Liquid Automatic Particle Counters using "AC" Fine Test Dust (ANSI B93.28). The required counts shall be the average of not less than three consecutive counts. Samples may be taken from the reservoir or upstream of the filter(s) used for cleanup.
      (2) Use any combination of on-the-forklift and off-the-forklift filtration for cleanup.
      (3) Connect the counter sensor to the system, using a bypass line and a branch line. The bypass line shall connect to the system, have a flow rate not less than 5 times the sensor flow rate, and may return fluid directly to the reservoir. The branch line to the sensor shall tee into the bypass line and shall be not more than 12 inches in length. Any flow restriction devices shall be placed downstream of the particle counter sensor. Start engine and operate at not less than 1000 rpm. Record particle counts.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-53121	2. DOCUMENT DATE (YYMMDD) 910930

3. DOCUMENT TITLE
Truck, Lift, Fork, Clean Burning Diesel, Front/Side Loading, 6000 Pound Capacity

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Gerry Harlow	b. TELEPHONE (Include Area Code) (1) Commercial (703) 664-1806 (2) AUTOVON 354-1806
c. ADDRESS (Include Zip Code Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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